

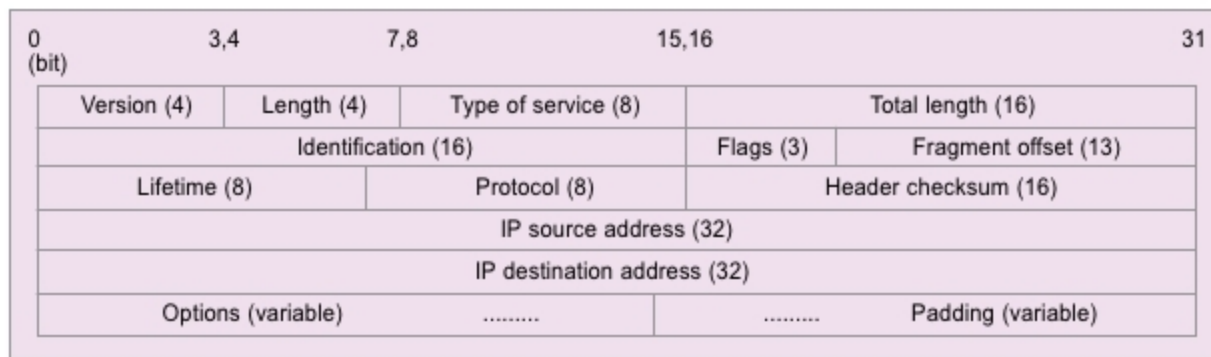


Fig 4: IPv4 headers

data is fragmented into three packets at TCP layer as: data pack 1, 2 and 3. Each data pack is added with TCP headers to make TCP packets.

TCP Packet = Data Pack + TCP headers (added at TCP layer)

Now, TCP packet is sent to the IP layer. With each of the TCP packets, IP layer adds IP headers making an IP packet (frame or datagram).

IP Packet = TCP packet + IP headers (added by IP layer)

Then, IP datagram is released to the physical layer.

IP header descriptions

The current version of IP that is in use mostly is version 4, denoted as IPv4 (though IPv6 is also now in use in many places). Formation of packets under IPv4 is made with headers as shown in Fig. 4. Various headers are described as next. Headers at each level is made up of 4 bytes or 32 bits.

Headers at the first level are:

- 4-bit version field, which indicates IP version used in the IP datagram. The most widely-used version is IPv4. IPv6 (IP version 6) has already emerged as a better alternative.

- 4-bit length field, which is also known as the Internet header length (IHL). It contains information about the number of 32-bit words in the header of the IP datagram. The minimum number of headers in any IP datagram is five 32-bit or 20-byte words. This happens when there is no other option or padding field.

In this case, the IHL field contains 0101. As the IHL field can have the maximum value as 1111 (= 15), maximum number of headers in any IP datagram is fifteen 32-bit or 60-byte words. As the number of headers are in multiples of 32-bit or 4-byte words, headers that an IP datagram can carry is one of the following sizes: 20, 24,

28, 32, 36, 40, 44, 48, 52, 56 or 60.

- 8-bit type of service (TOS) field, which indicates whether the IP datagram should receive any special type of service. Based on the indication about special service offers, priority or precedence is given to the particular packet(s) over other packet(s). This enables the routers to choose the appropriate path for the transport of the packet(s).

In the past, this field was largely unused. TOS field splits into two sub-fields: 5-bit D/T/R/C/R (low delay/high throughput/high reliability/low cost/reserved) and 3-bit precedence.

Precedence is absolute and not relative. An IP datagram for a regular packet with no special priority will have D/T/R/C/R sub-field set as 0000 and precedence sub-field set as 000. Consider a multimedia packet that may need a higher throughput and highest-level precedence. For a multimedia packet like that, IP datagram will set D/T/C/R sub-field as 0100 and precedence sub-field as 111.

TOS field is used by routers to choose the routing path. If a regular datagram that needs a normal service reaches a router, the router selects a normal route as per the outing strategy. But when a datagram requiring minimum cost reaches a router, the latter selects a path with minimum number of hops.

Similarly, for a datagram with higher reliability, the router selects a more reliable (lower BER) path. Likewise, for a datagram with higher throughput, a path with higher link capacity needs to be selected. The standard is already set for various services of different TOS fields.

- 16-bit total length (TL) field, which indicates the total number of bytes that the datagram carries including headers. Maximum number of



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